

DNS Service – Complete Detailed Explanation

1. What is DNS?

DNS stands for **Domain Name System**. It is like the **phonebook of the Internet**.

- Every device on the Internet uses **IP addresses** to communicate.
- Humans cannot remember long numbers easily, so we use **domain names** like www.google.com.
- DNS converts a **domain name** into its corresponding **IP address**, so your device can find the website.

Example:

www.google.com → 142.250.190.68

So when you type a website in your browser, DNS makes it possible for your device to locate the website.

2. How does the device find a DNS IP?

When you open a website in your browser:

1. **Step 1: Check Local DNS Cache**
 - Your device stores IP addresses of websites it has visited before in a **local cache**.
 - If the IP is already saved, your device uses it immediately without contacting a server.
 - This makes website loading faster.
 2. **Step 2: Send request to DNS server**
 - If the IP is not in cache, the device sends a **DNS query** to a DNS server.
 - This DNS server will look up the IP address for the domain.
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3. Which DNS server does the device contact?

The DNS server your device contacts depends on your **network configuration**. Usually:

Type of DNS	Description	Example
ISP DNS	Provided by your Internet Service Provider	Your home ISP server
Public DNS	Open for everyone, faster, sometimes more reliable	Google DNS 8.8.8.8, Cloudflare 1.1.1.1
Private DNS	Used in organizations, schools, or companies	Company DNS server

- Your device will use the DNS server that is **configured in network settings** (manually or automatically from DHCP).

4. DNS Query Flow (Step by Step)

When your device does not have the IP in cache, the DNS query follows multiple steps:

- 1. Device checks local cache**
 - If the IP is stored → use it
 - Else → move to next step
 - 2. Device sends query to recursive / ISP DNS server**
 - The ISP DNS tries to find the IP in its cache
 - If found → return IP to device
 - If not → forward the query to Root DNS server
 - 3. Root DNS server**
 - The root server does not have the exact IP but knows the **TLD server** for the domain.
 - Example: for www.google.com, the root server directs to .com TLD server
 - 4. TLD (Top Level Domain) DNS server**
 - The TLD server knows which **authoritative server** manages the domain
 - Example: .com TLD server will point to google.com authoritative server
 - 5. Authoritative DNS server**
 - This server has the **actual IP address** of the domain
 - It sends the IP back to the requesting DNS server → then back to the device
 - 6. Device receives IP**
 - The device stores the IP in **local cache** for future use
 - Now it can access the website quickly
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5. DNS Flow Diagram (Long Detail)

Your Device

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Local DNS Cache (check if IP saved)

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ISP / Recursive DNS Server (check cache)

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Root DNS Server (.com, .in, etc.)

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TLD DNS Server (.com, .in, .org)

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Authoritative DNS Server (contains actual IP)

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Return IP to Device

- Each step is a **layer in the DNS lookup process**
 - Local cache ensures **faster website loading next time**
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6. Example (Real Life)

- You type **www.google.com** in your browser
 - Your laptop checks local cache → not found
 - Device contacts ISP DNS → not found → Root DNS → **.com** TLD → Google authoritative DNS
 - Google authoritative DNS sends **142.250.190.68**
 - Laptop saves IP in cache → browser connects to website
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7. Key Points to Remember

1. **DNS = Internet phonebook** → domain name → IP
 2. **Local cache** → saves time for repeated visits
 3. **Types of DNS servers:** Root, TLD, Authoritative, Recursive/ISP
 4. **Public DNS** → faster, secure
 5. **DNS query flow ensures correct IP is returned**
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8. Commands to Check DNS (Practical)

- `nslookup www.google.com` → shows IP address
 - `dig www.google.com` → shows full DNS query details
 - `ping www.google.com` → shows IP and connectivity
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9. Summary (Long Version)

1. Device first checks **local cache** → If IP found → uses it
2. Else → sends query to **ISP / Recursive DNS**
3. ISP DNS → forwards to **Root DNS** if not found
4. Root DNS → directs to **TLD DNS**
5. TLD DNS → directs to **Authoritative DNS**
6. Authoritative DNS → sends **IP** back to device
7. Device stores IP in **local cache** → website opens

✓ **Simple answer:** Device first checks cache, then ISP / Public DNS → Root → TLD → Authoritative DNS → gets IP → saves in cache → opens website
